CCP

CCP[30]

Core Idea:

- $CCP(x_j) = P\big(\text{Meaning}(x_{1:j}) | x_{<j}, \text{ClaimType}(x_{1:j})\big).$

$$= \frac{P(\text{Meaning}(x_{1:j}), \text{ClaimType}(x_{1:j}) \mid x_{<j})}{P(\text{ClaimType}(x_{1:j}) \mid x_{<j})}. \longrightarrow P(Meaning(x_{1:j})|x_{<j})$$

- $P(Meaning(x_{1:j})|x_{<j}) = \sum_{x_j^k \in M(x_j)} P(x_j^k|x_{<j})$
- $P(ClaimType(x_{1:j})) = \sum_{x_j^l \in CT(x_j)} P(x_j^l|x_{<j})$
- $CCP(x_j) = \frac{\sum_{x_j^k \in M(x_j)} P(x_j^k \mid x_{<j})}{\sum_{x_j^l \in CT(x_j)} P(x_j^l \mid x_{<j})}.$
- $CCP_{claim}(C) = 1 - \prod_{j \in C} CCP_{word}(x_j).$

Task：Fact-Checking

Sentence:
She attended the School of Design, where she earned a Bachelor of Fine Arts Degree in painting.

Fact: She earned a Bachelor of Fine Arts Degree in painting.

She attended the School of Design, where she earned a Bachelor of Fine Arts Degree in ...

- painting 49%
- art 3%
- acting 7%
- sculpture 2%
- 1977 39%

NLI(Bachelor of Fine Arts Degree in painting, Bachelor of Fine Arts Degree in painting) = entail
NLI(Bachelor of Fine Arts Degree in painting, Bachelor of Fine Arts Degree in art) = entail
NLI(Bachelor of Fine Arts Degree in painting, Bachelor of Fine Arts Degree in acting) = contra
NLI(Bachelor of Fine Arts Degree in painting, Bachelor of Fine Arts Degree in sculpture) = contra
NLI(Bachelor of Fine Arts Degree in painting, Bachelor of Fine Arts Degree in 1977) = neutral

top-K words (K=10)

CCP_{word}(painting) = (P(painting) + P(art)) / (P(painting) + P(art) + P(acting) + P(sculpture))
= 0.52 / 0.61 = 0.85

CCP_{word}(of) = CCP_{word}(in) = 1 (functional words)

CCP_{claim}(She earned a Bachelor of Fine Arts Degree in painting.) =
-(CCP_{word}(Bachelor) * CCP_{word}(of) * CCP_{word}(Fine) * CCP_{word}(Arts) * CCP_{word}(Degree) * CCP_{word}(in) * CCP_{word}(painting))

[30] Fact-Checking the Output of Large Language Models via Token-Level Uncertainty Quantification, ACL 2024

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- **Task Type:** Open-ended generation

- **Introduction:** CCP re-aggregates token logits based on semantics and claim type. It obtains a word-level CCP score by dividing the sum of probabilities of synonyms by the sum of probabilities of words of the same type. Then, it calculates the claim-level score using the token-level score. This method makes fact-checking independent of changes in surface word form or sentence structure.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Token-level	White-box	Post-hoc	Medium	Ranking

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